SOUTH LAKE HOME EDUCATORS



Lesson 5

Programming Basics IV: Enemy Ranks

Semester 1, Week 5

At a Glance

Lesson 5 objectives & materials

By the end of today, students will:

- Explain what a loop is and why it avoids repeating code
- Explain why one enemy's code already scales to many
- Explain Scratch's clone-identity limitation and its fix
- Turn a single enemy into a row of independent enemies

What We're Using Today

- Demo laptop with all 8 engines verified
- GameMaker, for the live lecture demo
- Engine-specific Student Handouts, one per student
- No new assets this week - everything reuses Lesson 3 and 4's art and sound
- Your own [YourName]Game project from Lesson 4 - not a new one

IGDA framework tie-in: Programming - loops/iteration, code reuse and object-oriented thinking (introductory).

Today's Agenda



0:00 - 0:05	Welcome back, Lesson 4 recap, quick share-out
0:05 - 0:25	Loops: Doing Something Many Times Without Repeating Yourself
0:25 - 0:45	Scaling Behavior: Why One Enemy's Code Already Works for Many
0:45 - 0:50	A Scratch-Specific Wrinkle: Clones Have No ID
0:50 - 1:00	Introduce lab exercise, hand out engine-specific handouts
LAB TIME	***— Enemy Ranks —***
1:00 - 1:50	Turn your single enemy into a row of enemies
1:50 - 2:00	Exit check: quick share-out



Welcome Back

Lesson 4 recap and a quick share of any working combat mechanics.

5 min

Welcome Back

5 min

Quick recap of Lesson 4:

- Collision detection - a bullet touching an enemy destroys both, cleanly and exactly once
- Everyone got one working enemy that reacts correctly to being hit
- Today: one enemy becomes many, using a loop instead of repeating yourself

Open your own project from last week today, not a new one - and no new assets needed either, everything builds on Lesson 4's enemy and collision detection.



Loops

Doing something many times without writing it out many times.

20 min

Loops

20 min

A loop runs the same block of instructions a set number of times, instead of writing it out repeatedly by hand.

Change something about how enemies look later? Edit one loop, not five copy-pasted blocks.

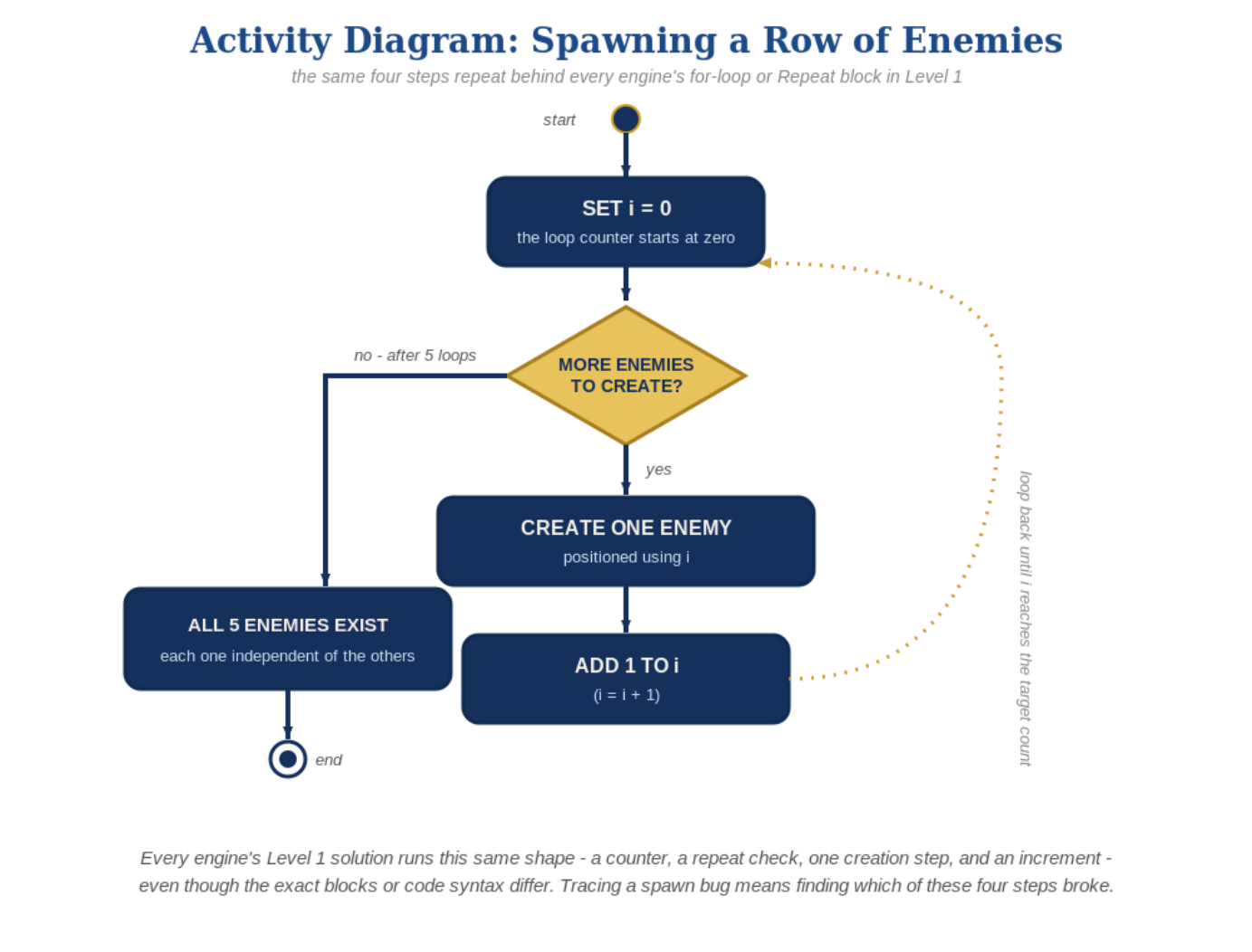
Same underlying idea, different names, across every engine:

Godot / Unity / GameMaker	`for (i = 0; i < 5; i++) { ... }`
GDevelop	`"Repeat 5 times" event`
Unreal (Blueprint)	`For Loop node`
TIC-80	`for i=0,4 do ... end`



Discussion: where else might a loop save you from repeating yourself in a game?

The Spawn Loop, Step by Step



Same counter, same repeat check, same creation step, same increment - whatever the exact blocks or syntax look like in your engine.



Scaling Behavior

Why last week's one-enemy code already handles a whole row of them.

20 min

Scaling Behavior

20 min

Last week's hit-response code doesn't change at all - it just runs once per enemy, automatically, because every instance carries its own independent copy of it.



This is why the Structure principle from Lesson 4 mattered: keeping an object's own response to being hit inside that object's own script is exactly what lets it scale to any number of copies for free.



Live demo: shoot enemies out of order - the third one, then the first, then the last. Each reacts completely independently, with nothing extra written to make that true.



A Scratch-Specific Wrinkle

Clones have no ID - so this week the hit-check moves to the enemy.

5 min

Clones Have No ID

5 min

Unlike instances in every other engine here, Scratch clones have no unique ID number to check against.

Other engines

A Bullet can ask "which specific Enemy am I touching?" and act on just that one.

Scratch

A Bullet clone can only ask "am I touching any Enemy?" - not which one.

Fix: move the check onto each Enemy clone's own script - each one asks "am I touching a Bullet?" about itself, which works correctly no matter how many enemies exist. This is a real change from last week's code, not just an addition.



Today's Exercise & Lab

Enemy Ranks - turning one enemy into a whole row of them.

10 min

Enemy Ranks

10 min

Several enemies appear in a row, created by a loop. Shooting any one destroys just that one - built on top of your existing project.

Level 1 - Minimal

A row of enemies appears via a loop; any one can be destroyed.

MVP - a complete finish for today.

Level 2 - Complete

Every enemy is independently destructible with no errors.

Everything works end to end.

Level 3 - Refined

Sound already works for free - add row-to-row visual variety.

Ready to share.



Reminder: delete the single enemy you manually placed last week before testing your new loop-spawned row, or you'll end up with one extra enemy that doesn't belong.

Lab Time Notes



- Scratch needs real hands-on support this week - the touching-check has to move from Bullet to Enemy, a genuine restructure. Expect this cluster to take longer.
- Most common bug elsewhere: forgetting to delete last week's manually-placed single enemy, leaving one extra enemy that doesn't belong to the row.
- Second most common: all spawned enemies stacking on top of each other - almost always means the loop's index variable isn't used in the position math.
- Students who finish Level 2 early should be pointed at the Level 3 ideas on their handout - not given new unrelated work.



Great Work Today!

Next week: a win condition - clearing all enemies completes the Level 1 MVP milestone.

Keep polishing your enemy row at home if you'd like - try one of the Level 3 ideas if you haven't already.